

Project 2: Finite Volume

AE 623, Computational Fluid Dynamics II, Winter 2023

Due: February 17, 11:59pm, electronically via Canvas

1 Problem Description

1.1 Introduction

In this project you will write a finite-volume solver to perform simulations of inviscid, compressible flow over the multi-element airfoil from the last project. You should use the meshes that you generated in that project for your runs. The “coarse” mesh refers to your locally-refined mesh of approximately 2000 cells.

1.2 Governing Equations

The governing equations are compressible Euler with ratio of specific heats $\gamma = 1.4$. You should implement your finite volume method using the conservative state vector $\mathbf{u} = [\rho, \rho u, \rho v, \rho E]^T$. Use convenient units, in which the gas constant is $R = 1$, and the free-stream state is given by

$$\mathbf{u}_\infty = [\rho, \rho u, \rho v, \rho E]^T = \left[1, M_\infty \cos \alpha, M_\infty \sin \alpha, \frac{1}{(\gamma - 1)\gamma} + \frac{M_\infty^2}{2} \right]^T, \quad (1)$$

where M_∞ is the free-stream Mach number, and α is the angle of attack. The airfoil elements are solid inviscid walls, while the farfield is a full-state boundary, where the free-stream state is specified outside the domain. All boundary conditions should be implemented according to the methods described in the notes.

1.3 Flux Functions

Implement three Riemann Solvers: Roe, Rusanov, and one of your choice (e.g. HLLE, HLLC, Osher, flux vector splitting, etc. – see *Riemann Solvers and Numerical Methods for Fluid Dynamics* by E. F. Toro). For the Roe flux, use an entropy fix with $\epsilon = 0.1c$.

1.4 Second Order and Limiters

Implement both first-order and second-order finite-volume methods. For second-order, consider three possibilities for limiting: none, Barth-Jespersen (BJ), and one of the maximum-principle (MP) region limiters (e.g. LCD, Durlofsky, MLG). The choice of gradient calculation for the BJ limiter is up to you.

1.5 Flow conditions

Consider two flow conditions:

1. **Subsonic:** $M_\infty = 0.25$, $\alpha = 8^\circ$
2. **Transonic:** $M_\infty = 0.5$, $\alpha = 8^\circ$

Figure 1 shows the expected Mach number contours for both of these cases.

1.6 Time Stepping and Convergence

Use SSP-RK2 with local time stepping to converge the flow to steady state. When starting a fresh run, initialize the state to free-stream. For uniform-refinement studies, you can use the solution of a coarser mesh to initialize the state on a finer mesh. For the second-order solutions, you can use

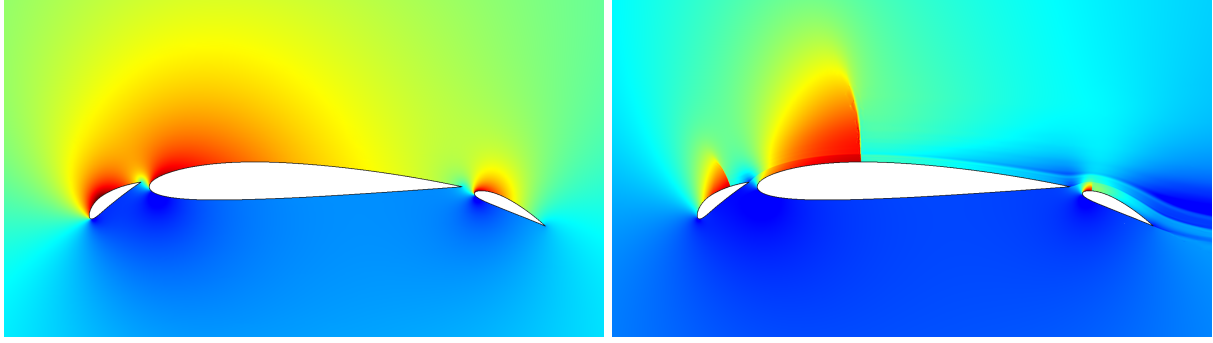


Figure 1: Mach number contours for the subsonic (range is 0 to 0.7) and transonic (range is 0 to 1.7) flow conditions.

the first-order solutions as initial guesses. In all runs, monitor the discrete L_1 norm of the entire residual vector, taken together over all components,

$$|\mathbf{R}|_{L_1} = \sum_{\text{entries}} |R_i|. \quad (2)$$

Consider a solution converged when $|\mathbf{R}|_{L_1} < 10^{-5}$.

1.7 Outputs

In addition to flow-field plots of Mach number, of interest are two scalar outputs: the lift and drag coefficients:

$$c_\ell = \frac{L'}{\frac{1}{2}\rho_\infty |\vec{v}_\infty|^2 c}, \quad c_d = \frac{D'}{\frac{1}{2}\rho_\infty |\vec{v}_\infty|^2 c} \quad (3)$$

where $c = 1$ is the reference chord length, $\vec{v}_\infty$ is the free-stream velocity, and L' and D' are, respectively, the sectional lift and drag forces on the airfoil, summed together over all three elements. Drag is the component of the force aligned with the free-stream velocity, while lift is the orthogonal component. Compute the forces by integrating the pressure over the wall boundaries. Finally, you are also asked to compute the pressure-coefficient distribution over the airfoil (all elements). The pressure coefficient is defined as

$$c_p = \frac{p - p_\infty}{\frac{1}{2}\rho_\infty |\vec{v}_\infty|^2} \quad (4)$$

Note that by aerodynamic convention, we typically plot $-c_p$ versus x .

1.8 Adaptation

Apply solution-based mesh adaptation from Project 1 to improve solution quality. A reasonable way to measure error is to look at jumps in the solution, e.g. the Mach number, between cells. For each interior edge e , we can define an adaptive indicator:

$$\text{interior: } \epsilon_e = |M_{k+} - M_{k-}| h_e. \quad (5)$$

In this formula, M_{k+} and M_{k-} are the cell-average Mach numbers on the two cells adjacent to edge e , and h_e is the length of edge e .

You can assume that the error indicator on the farfield boundary edges is zero. On the airfoil boundaries, define the error indicator by

$$\text{wall: } \epsilon_e = |M_k^\perp| h_e, \quad (6)$$

where $M_k^\perp$ is the Mach number associated with the cell's edge-normal velocity component.

After calculating the error indicators ϵ_e over all edges (interior and boundary), sort the indicators and flag a fraction $f = .03$ of edges with the highest error for refinement. Smooth the refinement pattern by looping over all cells: if a cell has *any* of its edges flagged for refinement, then flag *all* of its edges for refinement. This will increase the total number of edges for refinement. Once edges are flagged as described, refine all cells adjacent to flagged edges. At each adaptive iteration, transfer the solution to the new mesh to provide a good initial guess for the next solve.

2 Tasks

Address the following tasks in your report.

1. **[10%]** Flux function implementation and testing. State the flux functions that you chose, describe how you verified that your flux functions are correct, and present results from one or two of your verification tests.
2. **[15%]** First-order solver. Describe your algorithm and include a c_p plot for the subsonic condition, using your choice of mesh and flux function.
3. **[20%]** Second-order solver and limiters. Describe your algorithm and explain your gradient calculation and limiter. Include a c_p plot for the same case as the first-order solver – you can overlay your c_p plots.
4. **[20%]** Convergence studies on the subsonic flow condition. You do not have to run all of the possible combinations of solvers, fluxes, limiters, grids. Instead, your goal should be to run enough simulations to answer the following questions:
 - How do the flux functions differ in dissipation?
 - Is the first/second-order method actually first/second order accurate?
 - Do limiters impact the convergence rate?

Use the lift and drag coefficients as the scalar outputs in your studies – you do not need to show both for every study.

5. **[15%]** Transonic runs. On at least two meshes, medium/fine, run the transonic flow condition with both your first and second order solvers. Choose one flux function and limiter – ideally the most accurate. Report the lift/drag coefficients, and show c_p plots and Mach number contours. Discuss your results.
6. **[10%]** Adaptive run. Starting with your coarse mesh, run your first and second-order solvers for at least five adaptive iterations, using the transonic flow condition. Using your finest-mesh results, provide a c_p plot and lift/drag coefficients. Compare these to the uniform-refinement results.

Note that accompanying this assignment are several codes that you may find useful, including a few flux function implementations and plotting functions.

3 Deliverables

Each group should turn in, electronically via Canvas,

1. A technical report, **12-pages maximum** for this project, as a `.pdf` file that describes your methods and results, and that answers the questions. The report should be professional, complete, and concise. **10%** of your grade will be determined by the presentation: neatness, labeling of axes, spelling, etc.
2. Documented source files of all codes you wrote for this project, as one `.zip` archive. You should provide all files necessary to run your program(s).

This is a group assignment. At the end of the report, summarize the contributions of each group member to the method development, code, results, and report.